Round Robin, Multilevel Queue & Multilevel Feedback Queue

Time Quantum

What is Time Quantum?
The maximum amount of CPU time allocated to a process during one turn.
Example:
Time Quantum = 3 ms

P1 → 3 ms
P2 → 3 ms
P3 → 3 ms
P4 → 3 ms
P1 → 3 ms
...

Round Robin Example

Process	BT
P1	5
P2	3
P3	4

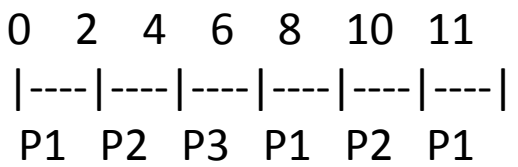
Round Robin Numerical 1

Process	AT	BT
P1	0	5
P2	0	4
P3	0	2

Time Quantum
q=2

Process	CT	AT	TAT
P1	11	0	11
P2	10	0	10
P3	6	0	6

Process	TAT	BT	WT
P1	11	5	6
P2	10	4	6
P3	6	2	4



Response Time in RR

$$RT = First\ CPU\ Start - AT$$

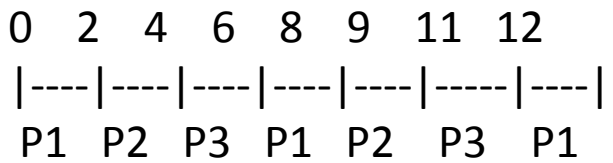
Process	AT	First Start	RT
P1	0	0	0
P2	0	2	2
P3	0	4	4

$$AverageResponseTimeART = \frac{0+2+4}{3} \boxed{ART = 2}$$

Round Robin with Arrival Times

Process	AT	BT
P1	0	5
P2	1	3
P3	2	4

Time Quantum
q=2


0 2 4 6 8 9 11 12
P1 P2 P3 P1 P2 P3 P1

RR Numerical Practice

Process	AT	BT
P1	0	6
P2	1	4
P3	2	5
P4	3	2

q=2

Algorithm	Type	Selection Basis
FCFS	Non-preemptive	Arrival order
SJF	Non-preemptive	Shortest BT
SRTF	Preemptive	Shortest remaining BT
Priority	Either	Priority
Round Robin	Preemptive	Time quantum